

TraceFall: Team Briefing and Presentation Notes

Smart India Hackathon 2026 · Problem Statement SIH26183

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If something goes wrong on stage

The 60-second version

Read this page first. If you remember nothing else, remember this page.

What we built. A web application for cybercrime police. An investigator types in the wallet address a fraud victim gave them. About ninety seconds later the screen shows where the money went, which service received it, how risky the trail looks, and a report they can attach to the case file.

Why it matters. When someone is cheated out of money through a bank transfer, the account number tells the police which bank to write to. When someone is cheated in crypto, the wallet address tells them nothing at all. No bank name, no branch, no country. So the investigator opens a public blockchain explorer, clicks through transactions by hand, and usually gives up after two hops. That takes six to twelve hours, and by then the money is gone.

Our one-line pitch:

TraceFall answers one question for an investigator: where did the money go, and which exchange do I send the freeze request to?

The three things that make us different. Say these in every conversation.

1. **Every single number on screen can be traced back to a real blockchain transaction.** Click any finding and it opens up into the transactions behind it, and finally into the raw data we saved, with a timestamp and a fingerprint. Nothing is a black box.
2. **We never guess and present it as fact.** Every claim about who owns an address carries one of three labels: CONFIRMED, PROBABLE, or UNATTRIBUTED. These are never mixed up or blurred. The database itself refuses to store anything else.
3. **When we do not know something, the system says so.** It shows which branches it could not follow and why. A tool that quietly gives you a smaller answer is worse than one that admits the gap.

The number to know: the demo runs on real blockchain data, saved in advance, with the laptop's internet switched off. Nothing is fake and nothing is live.

The problem

What SIH asked for

Problem statement SIH26183, Ministry of Home Affairs.

Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics.

In plain words: a citizen gets scammed and reports a wallet address. Find out where that money ended up, and which company now holds it, fast enough that the police can still do something about it.

Why this is genuinely hard

A wallet address is just a long string of letters and numbers. It carries no name, no bank, no country, nothing. Anyone can make one in two seconds, for free, with no ID.

Scammers know this, so they never keep the money in one place. They split it, move it through five or ten wallets, merge it back together, and only then send it to an exchange where it can be turned into cash. Following that by hand is slow and error-prone.

And there is a deadline. Once the money reaches an exchange and gets withdrawn, it is usually gone for good. The window is hours, not weeks.

The one thing the police actually need

Here is the key insight behind our whole product.

You cannot arrest a wallet address. But you **can** write to a company. Every exchange, like Binance or CoinDCX, is a real registered business with real offices and real compliance officers. They know who their customers are, because Indian and international rules force them to collect ID.

So the moment you can say “this money reached exchange X”, the investigation stops being a technology problem and becomes a normal legal process. You send a freeze request, they freeze the account, they tell you who owns it.

Getting the investigator to that moment is the entire job of TraceFall.

How TraceFall works

Nine steps run automatically after the investigator submits an address. This is what the progress screen shows while it works.

#	Step	In plain language
1	Retrieve	Download every transaction this address was ever part of
2	Normalize	Convert different data formats into one clean, common shape
3	Enrich	Work out what kind of address it is and how it behaves
4	Trace	Follow the money forwards, hop by hop
5	Graph	Turn those hops into a picture you can look at
6	Patterns	Spot shapes that look like laundering
7	Attribute	Work out who owns the addresses the money reached
8	Risk	Score how serious this looks, and show every reason
9	Report	Produce a PDF for the case file

The two steps worth explaining properly

Judges will ask about these two. Everyone on the team should be able to explain both.

TRACING: HOW WE FOLLOW THE MONEY

Money does not move in a straight line. One wallet sends to five wallets, those send to twenty more. If you follow everything, you get a useless mess of thousands of addresses.

So we follow **the victim's money specifically**, not all money. Every hop, we track what share of the funds came from the original suspect address. If a branch carries less than one percent of it, we stop following that branch, and we say on screen that we stopped and why.

We stop for six clearly-named reasons, and the screen always tells you which one:

- **MAX_DEPTH** we reached our five-hop limit
- **BELOW_THRESHOLD** too little of the victim's money went this way to matter
- **SERVICE_BOUNDARY** the money reached an exchange, so this is the answer
- **NO_OUTFLOW** the money arrived here and never left
- **EDGE_BUDGET** the trail got too wide to follow completely
- **DATA_UNAVAILABLE** we could not fetch this hop's data

That last distinction matters a lot. **"The money stopped here" and "we could not look here" are completely different findings**, and most tools blur them together. We never do.

One more rule: we stop when we reach an exchange, on purpose. Everything past an exchange's main wallet is other customers' money, and following it would mean pointing at innocent people.

ATTRIBUTION: HOW WE SAY WHO OWNS AN ADDRESS

This is the part that answers the problem statement, so get it right.

We put every address into exactly one of three buckets, and we never mix them:

Label	What it means	Where it comes from
CONFIRMED	A named, dated, public source says this address belongs to this company	Published lists we have loaded, such as the US Treasury sanctions list, or an exchange's own published wallet disclosure
PROBABLE	The address behaves exactly like one belonging to this company	Our own rules, always shown with a confidence figure and the reasons behind it
UNATTRIBUTED	We do not know	We say so plainly, and we say what would be needed to find out

The database physically refuses to store any other value. This is not a rule we remember to follow, it is enforced by the system.

Why this matters so much. A freeze request sent to the wrong company is worse than no freeze request at all. It wastes police time, it damages a real business, and it warns the criminal. So we would rather say “we do not know” than guess.

What we deliberately did not use: AI

We ship **no machine learning model at all**, and this is a deliberate design decision, not something we ran out of time for.

The reason is simple. An investigator has to explain their findings in a courtroom. “The model gave it a score of 0.87” cannot be explained. “This address received money from a sanctioned wallet on this date, in this transaction” can be.

So every number we show is produced by a rule a human can read, in a config file anyone can open. We wrote the decision down formally before building it.

If a judge asks where the AI is, that is a **good** question to get. See the Q&A section.

What the demo shows

Four real wallet addresses. All four are on the US Treasury's published sanctions list, which means an actual government has publicly named them as connected to crime. We did not invent them and we did not pick them because they gave us flattering results. We chose them by shape before we ran a single one.

#	What it demonstrates	Time	Result on screen
1	The full trace. Money moving across many hops	70 sec	26 addresses, 30 connections, reaches a second sanctioned address
2	Smart pruning. Money split many ways, most of it too small to matter	2 sec	2 addresses, and it explains why the rest were dropped
3	Funds have not moved. The money arrived somewhere and stayed	2 sec	2 addresses, ends with NO_OUTFLOW
4	Honest failure. Too little activity to identify anything	30 sec	Root address reported UNATTRIBUTED

Case 4 is the most important one. Do not skip it for time.

It sounds strange to show the judges a case where the system does not identify anything. Here is why it is our strongest moment.

We measured our identification method properly, against 2.3 million addresses that Binance publishes as its own. When our system does identify an address, it is right **98.9 percent of the time**. But it can only identify about **19 percent** of the addresses it sees. The other four in five simply do not have enough history to be classified.

We show both numbers, always, together. Anyone can build something that answers every time if it is allowed to guess. We chose the other trade: **a missed identification costs an investigator some time, a wrong one sends a legal notice to the wrong company.**

Showing this case first, before a judge finds it, turns a limitation into evidence that we know what we built.

About “PARTIAL”

Case 1 finishes with the word PARTIAL on screen, and two branches marked as unavailable. **Do not apologise for this.** Point at it.

It means two hops were not in our saved data, and the system said so instead of quietly showing a smaller trail. A real investigation always has branches nobody could follow. A demo where every single branch resolves perfectly would be the suspicious one.

Why this is strong

Points to make, roughly in order of how much they impress people.

1. It actually works, end to end, offline. Real blockchain data, saved in advance. The laptop's network is off during the demo and the screen says "cached snapshot" on every page. Nothing is mocked, nothing is a video.

2. Every claim is checkable. 1,959 pieces of stored evidence in the demo run, each with a hash and the exact time we retrieved it. Click any number, get the transactions. Click a transaction, get the raw data.

3. We measured ourselves honestly. Accuracy of 0.989 with recall of 0.186, both stated. Timings of 70, 30, 2 and 2 seconds against a 120-second target, measured, not estimated.

4. It is properly built software, not hackathon glue. 539 backend tests, 60 frontend tests, and 5 tests that drive a real browser. Over 97 percent test coverage on the three engines that matter. All code checks pass cleanly.

5. It cannot silently rot. There is a check that replays all four demo cases offline and fails loudly if anything is missing, and a locked "golden" result file that catches any number changing unexpectedly.

6. We built past the problem statement. Two extras nobody asked for:

- **Cross-case linking.** If two victims report addresses that both lead to the same wallet, the system spots it. Ten victims funnelling into one address is one freeze request instead of ten, and it turns ten small cases into one big one.
- **Freeze request drafting.** The system writes the actual letter to the exchange, and it carries the confidence level into the wording. A PROBABLE identification never reads like a confirmed fact.

7. Privacy and security by design. No victim personal data is ever stored, only reference numbers. No secret keys in the code. The system is read-only against the blockchain, it can never move anyone's money. Fresh installs need no API keys at all.

The technology, in one table

Layer	What we used
Frontend	React, TypeScript, Tailwind, Cytoscape.js for the graph
Backend	Python 3.12, FastAPI
Database	PostgreSQL 16
Cache and queue	Redis 7
Blockchain data	TronGrid and TronScan for TRON, Etherscan and Blockscout for Ethereum
Graph analysis	NetworkX
Reports	ReportLab for PDF
Deployment	Docker Compose, one command

Chains supported: **TRON and Ethereum, USDT first**. That is a deliberate choice, not a limitation we ran into. Most Indian crypto fraud money moves as USDT on TRON, because the fees are almost nothing. We would rather do two chains properly than six badly.

What we deliberately do not claim

Learn this table. If a judge pushes on any of these, the honest answer is the impressive one.

We can say	We must never say
Traces fund flows across many hops on TRON and Ethereum	Traces across all blockchains
Identifies addresses linked to known exchanges, with confidence levels and evidence	Identifies which exchange the criminal used
Detects patterns consistent with laundering	Detects money laundering
Produces an investigative risk score with every signal shown	Calculates the probability of fraud
Cuts a multi-hour manual trace to about ninety seconds	Solves crypto fraud
Generates an investigation report for the case file	Generates court-admissible evidence
Everything is deterministic and inspectable, no model ships	AI-powered fraud detection platform
98.9 percent precision at 18.6 percent recall, both stated	98.9 percent accurate
About ninety seconds against the saved snapshot	Ninety seconds for any address, live
Designed for integration with systems like NCRP	Integrated with NCRP
Shows where blockchain tracing ends and legal process begins	Recovers stolen funds

This is not us being modest. Any judge who knows the field will test these claims. A team that has already drawn the line itself sounds far more credible than one that has to back down under questioning.

How to present it

Six minutes of demo, then questions. Assign these roles in advance and rehearse the handoffs.

- **Speaker** does all the talking.
- **Driver** operates the laptop and never talks.
- **Backup** watches the clock and handles questions during the demo.

Running order

1. The problem, 45 seconds. Say this before touching the laptop.

When a citizen reports a bank fraud, the account number tells you which bank to write to. When the same citizen reports a crypto fraud, the wallet address they hand over tells you nothing. No bank, no branch, no country. So the investigator opens a block explorer, clicks through transactions by hand, and gives up after two hops. Six to twelve hours, usually ending in “unknown”, long after the money has been withdrawn.

2. Start the trace, 45 seconds. New case, paste address 1, start. While it validates:

This is a real address, publicly designated by the US Treasury. It is on a government list you can go and check. We did not invent it, and we did not pick it because it gave a good answer. We picked four by shape before running any of them.

Then point at the “cached snapshot” banner:

Everything here is real blockchain data, frozen in time. The system is running with no network at all. It says so on every page, because a demo that quietly pretends to be live is the first thing I would distrust.

3. The pipeline, 30 seconds. The progress screen names each stage as it runs.

Nine stages. This one takes about seventy seconds, because the money runs into addresses with ten and twenty thousand transactions each. That volume is itself part of the finding. The quieter addresses finish in two seconds.

4. The findings, 3 minutes. Walk the tabs in this order.

- **Overview:** the plain answer. Where the money went, how much is accounted for.
- **Graph:** every dot is an address, every line a real transaction. Colour shows risk, border style shows how confident we are about ownership. There is a text list next to it saying the same thing, because a picture is not evidence and not everyone can read one.
- **Attribution:** this is the answer to the problem statement. Three labels, never mixed. Open one row so the evidence is on screen: the source, its date, the transactions.
- **Patterns:** each finding carries its own note about how it could be wrong, in the same panel. Most of these are on busy service addresses where this behaviour is completely normal, and the note says exactly that.
- **Risk:** zero to a hundred, every contributing signal listed with its weight. There is a line underneath saying this is not a probability of fraud, because it is not one.
- **Evidence:** generate the PDF, show its fingerprint, download it.

5. The address it cannot identify, 60 seconds. Run case 4. This is the most important minute. Use the wording from the “What the demo shows” section above.

6. Close, 20 seconds.

Blockchain tracing ends where legal process begins. Our job is to get the investigator to that boundary in ninety seconds instead of a day, with an evidence trail that survives being questioned, and to be honest about exactly where it stops.

Things not to do

- Do not say “ninety seconds for any address, live”. Say “against this saved snapshot”.
- Do not apologise for PARTIAL, for pruned branches, or for the unidentified case. Point at them. They are the product.
- Do not say “AI” or “machine learning” about anything we built. We ship neither, on purpose.
- Do not promise a feature we have not built. “Not yet, and here is why we prioritised differently” is a better answer than a bluff.

- If you do not know an answer, say “I do not know, I will find out”. Nobody loses marks for that. People lose marks for confident wrong answers.
-

Questions the judges will ask

Short answers, in plain language. Practise saying these out loud.

"So what does your project actually do?"

A fraud victim gives the police a crypto wallet address. Our system automatically follows the money across the blockchain, works out which company received it, scores how suspicious the trail looks, and produces a report. What takes an investigator a full day takes us about ninety seconds.

"How is this different from just using a blockchain explorer?"

An explorer shows you one address at a time and you click through the rest by hand. We follow the money automatically across multiple hops, work out how much of the victim's money went down each path, identify who owns the endpoints, and write the report. An explorer is a dictionary. We are a translator.

"Where is the AI in this?"

There is none, and that is deliberate. We wrote the decision down before we built it. An investigator has to defend their findings in court, and "the model scored it 0.87" cannot be defended. Every number we show comes from a rule you can open and read. We had a machine learning component planned, we scoped it, and we cut it because the plain rule already answered the question and a model would have made it harder to explain.

"How accurate is it?"

When we identify an address, we are right 98.9 percent of the time. We tested that against 2.3 million addresses Binance publishes as its own. But we can only identify about 19 percent of addresses in the first place, because the rest do not have enough transaction history. We always give both numbers. The second one is the honest one and it never travels without the first.

"Can you tell me who the person behind the wallet is?"

No, and nothing that only reads public blockchain data can. We never claim it. What we identify is the **company** that received the money, because that company has the customer's ID documents, and that is who the police send the legal request to.

"Which exchange did the money go to in this case?" (*our weakest spot, answer it head-on*)

In these four cases, none that we can name. The system names the receiving company whenever a published, first-party disclosure covers it, and it drafts the freeze request to that company. These four addresses happen to land on high-volume services that no exchange has publicly claimed as its own. We only add a company label when we can verify it from that company's own published wallet list, not from a scraped guess, because a freeze request sent to the wrong institution costs far more than a missed one. Expanding that list is a data problem, not a technical one, and it is the next thing we do.

"Is this real data or dummy data?"

Completely real, and frozen in time. Real addresses from a real government sanctions list, real transactions, real amounts. We saved the blockchain's responses in advance so the demo cannot be broken by bad conference wifi. The screen says "cached snapshot" on every page. We never generate fake blockchain data, at all.

"How long does it take on a real case, not a demo?"

The processing itself is fast. The limit is how quickly the free blockchain data providers will answer us. Without a paid key, that is around one request every two seconds, so a busy address takes longer. With a paid key it is much faster. Our own pipeline is not the bottleneck, someone else's free tier is.

"Can the police actually use this in court?"

It is an investigation tool, not a court exhibit, and we say that in the report itself. Every finding traces back to a transaction hash and the raw stored data with a timestamp, so it stands up to being questioned. Whether it is admissible is a decision for a court, and we do not answer that on the system's behalf.

"What if the criminal uses a different blockchain?"

Today we handle TRON and Ethereum, which is where the large majority of Indian crypto fraud money actually moves, mostly as USDT on TRON because the fees are tiny. Adding another chain means writing one new adapter file. We deliberately built it so that nothing in the tracing, graph, risk or report code has to change.

”What if they use a mixer, or a privacy coin?”

A mixer breaks the trail, and we say so on screen rather than pretending otherwise. That is a real limit of public blockchain analysis, not a gap in our build. Privacy coins like Monero cannot be traced this way at all, by design, and we do not claim to.

”Is it connected to NCRP or the police systems?”

No. We expose an integration-ready API, and that is all we claim. Saying we are integrated with a government system we have not connected to would be the easiest thing in this room to disprove.

”How do you know an address belongs to Binance and not somebody else?”

Because Binance publishes that list themselves, and we only use lists that a company publishes about its own wallets. We check every address is validly formatted, and then we look at how it actually behaves on the blockchain before we trust the label. We do not copy labels from other people’s databases.

”What stops you from being wrong and getting an innocent person’s account frozen?”

Three things. We never merge our confidence levels, so a “probable” identification is labelled probable everywhere including in the draft letter. We stop tracing at an exchange, because everything past it belongs to other customers. And we never label someone who paid **into** the scam wallet as a suspect, because they are almost certainly another victim.

”Why does one of your cases only show two addresses? Is it not working?”

It is working exactly as designed. That address sent money fifteen times to nine different places, but all except one were tiny fractions of the victim’s money. We stop following branches below one percent, and the screen tells you we did. Following everything would give you a thousand addresses and no answer.

”What does PARTIAL mean? Did it fail?”

No. It means two branches could not be followed because that data was not in our saved snapshot, and the system reported that instead of quietly showing you a smaller trail. A real trace always has branches nobody could follow. Hiding that is the failure.

”Who would run this? What is the deployment model?”

It runs inside a police network, on their own hardware, with one command. Case data never leaves. We store no victim personal information at all, only case reference numbers. The system is read-only against the blockchain, so it can never touch anyone’s funds.

”How much of this did you build yourselves?”

All of the logic. The tracing engine, the pattern detectors, the attribution rules, the risk scoring, the report generation, the entire interface. We use standard open-source libraries for the plumbing, a graph library and a PDF library, the way any production system does. The parts that make it work are ours.

”What is next?”

Three things. First, expand the list of exchanges we can name, by pulling in more companies’ own published wallet disclosures. Second, connect the identified company to India’s official FIU register, so the freeze request goes to a legally registered entity with a named compliance officer. Third, add more chains, which is one adapter file each.

Words you might be asked to explain

Keep these short if a judge asks. Do not lecture.

Wallet address. A long string of characters that acts like an account number on a blockchain. Free to create, no ID needed, no name attached.

Transaction hash. A unique fingerprint for a single transfer. It is how anyone can look up that exact transfer and check it themselves.

Hop. One move of money from one address to the next. A five-hop trace follows the money through five moves.

USDT. A cryptocurrency designed to always be worth one US dollar. It is what most fraud money is moved in, because its value does not swing around.

TRON. A blockchain network. It is popular for moving USDT because the fees are close to zero.

Exchange. A company where people buy and sell crypto for regular money. Binance, CoinDCX, WazirX. These are real registered businesses that hold customer ID.

Hot wallet. An exchange's main working wallet, where thousands of customers' funds sit together. We stop tracing here, because past this point the money belongs to other people.

Deposit address. The personal address an exchange gives one customer for paying money in. Money sitting here still belongs to that one customer, which is why identifying these is so useful.

OFAC sanctions list. A list published by the US Treasury naming people and wallet addresses connected to crime and terrorism. Public, dated, and checkable by anyone.

Taint. The share of money at some address that originally came from the suspect address. It is how we tell the victim's money apart from everything else flowing around.

Freeze request. The formal letter police send to an exchange asking them to lock an account and identify who owns it.

Pre-flight checklist

Run this before the room, in this order. Every step has broken at least once, which is why each one is a step and not an assumption.

```
docker compose up -d
docker compose exec api alembic upgrade head
docker compose exec api python -m app.cli load-labels
docker compose exec api python -m app.cli create-admin
python scripts/demo_fixtures.py --check
open http://localhost
```

The last check is the one that matters. It replays all four demo cases offline and fails loudly if anything is missing.

Then, before anyone is watching:

- **Sign in once** and leave the tab on the case list. The first analysis of a session is always the slowest.
- **Set the screen to 1366 x 768**, which is what most projectors give you. Check both light and dark themes and pick whichever reads better on the projector.
- **Switch the wifi off and reload the page**. The system makes no outside calls anyway, but proving it to yourself beforehand is what lets you say it flatly to a judge.
- **Have the backup recording open in another window**, one keystroke away.

If something goes wrong on stage

- **A case runs slowly**. Keep talking. The progress screen names each stage, so narrate what it is doing. That is a feature, not dead air.
- **A case fails outright**. Move to the next address. Do not debug in front of judges. Say “let me show you a different case” and carry on.
- **The whole stack will not start**. Switch to the recording immediately, and say so plainly. “Our environment is not cooperating, here is the recorded walkthrough” costs you far less than five minutes of silent troubleshooting.
- **A question stumps you**. “I do not know, I will find out.” Then move on.